Универзитет „Св. Кирил и Методиј“ во Скопје
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Дипломска работа

Способноста на мали јазични модели за промена на стил на македонски автори со примена на учење со поттикнување

[*Ментор: ...*] [*Кандидат: ...*]

Скопје, 2026

Abstract

[*Аспракт — written last (problem, method, the negative result with three or four numbers).*]

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Chapter 1

Introduction

1.1 Motivation

A poet's style is the part of a poem that survives a change of subject: the words they reach for, how long their lines run, whether they rhyme and how, which images return. Rewriting a poem so that its content stays and its voice becomes another poet's is a hard test of whether a language model has learned that layer of language rather than its surface. This is the task of *text style transfer*, here in its most demanding form: the "style" is not a register such as formal or informal, but one particular author.

Most work on style transfer and on language models in general is done in English, where data is plentiful. Macedonian is a low-resource language: it has few annotated corpora, few tools, and it is a small share of what general-purpose models are trained on. Poetry makes this worse. A poet writes a few hundred poems in a lifetime, often far fewer, so even a complete collection of an author's work is a small dataset by the standards of machine learning. The corpus this thesis is built on holds 1,239 poems and song lyrics by 38 Macedonian authors; the best represented author has 182 poems, the next two 160 and 134, and half of the authors have seventeen or fewer.

Large models can be prompted to write in a given style, but they are expensive to run and are not under the user's control. Small models of a few billion parameters can be trained and run on a single GPU, which makes them the practical choice for a language community without large computational resources. Whether they can learn a task as fine-grained as an individual poet's voice, from so little data, is the question this thesis sets out to answer.

1.2 Research question

The thesis asks:

> *Can a small language model, given only a source poem and the name of a Macedonian author, rewrite the poem in that author's style, and does reinforcement learning improve the result?*

Two parts of the question need making precise. "In that author's style" is measured in two ways that do not depend on each other: automatically, by stylometric features and by an author classifier that the training never sees, and by human judgement. "Reinforcement learning" is Group Relative Policy Optimization (GRPO) [*cite: Shao et al. 2024*], applied on top of supervised fine-tuning with a reward built from the same stylometric profile of each author.

1.3 Approach and main finding

The work reached its final form after several approaches that did not work. Replacing selected words of the source poem with words typical of the target author, inflected to fit, preserves the poem but changes little of its style. Asking small local models to fill such slots fails at Macedonian itself. What did work was a large model, Gemma 4 31B [*cite*], prompted with a detailed scaffold: a summary of the source poem, a prose description of the target author's style, a vocabulary palette and example lines, and explicit structural targets. That model became the *teacher*. Its outputs, about 115 per target author for ten target authors, are the training data for two *students* of similar size, Qwen3-4B [*cite*] and Gemma 4 E4B, which see only the source poem and the author's name. Each student is fine-tuned with LoRA [*cite: Hu et al. 2022*] on the teacher's poems and then trained further with GRPO.

The main finding is negative. The students learn the task's form: they write Macedonian, they write poems rather than prose, they change the source rather than copy it. They do not learn the voices. An independent author classifier recognises the intended author in 16.7% of the fine-tuned Qwen student's poems on the held-out test pairs, against 66.3% for real, unseen poems by the same authors; after GRPO the figure falls to 12.4%, even though the reward it optimises improves. [*Gemma student and teacher numbers on test; human ratings.*] The outputs that are identified belong almost entirely to the three authors with the most poems in the corpus. The thesis argues that the limit is the data rather than the method: a stronger student, a better-tuned reward and the teacher itself all run into the same small, uneven corpus.

1.4 Contributions

- A cleaned and part-of-speech-tagged corpus of Macedonian poetry, with poem-level identifiers, documented tagging corrections and a fixed train/validation/test split by author.
- Per-author stylometric profiles and a style-transfer reward built from them: part-of-speech distributions, lexical diversity, line and stanza structure, rhyme, characteristic vocabulary and an author classifier, with gates against copying the source and against degenerate output.
- A teacher–student pipeline that distils a large, heavily prompted model into two small students, followed by GRPO, run with matched settings for both students so that their results can be compared.
- An evaluation that separates real gains from reward gaming: the classifier used as a reward is never used to judge the outcome, a second classifier trained on different features is.
- A human evaluation of the teacher, both students and real poems on the same test pairs, with the real poems as a blind control.

1.5 Structure of the thesis

Chapter 2 reviews text style transfer, stylometry, knowledge distillation and reinforcement learning for language models. Chapter 3 describes the corpus and its preparation, and Chapter 4 the stylometric profiles and classifiers the rest of the work relies on. Chapter 5 summarises the earlier approaches and why they were abandoned. Chapter 6 presents the teacher, the students and

the reward; Chapter 7 the evaluation protocol. Chapter 8 reports the results, Chapter 9 discusses why the students fall short, and Chapter 10 concludes.

Chapter 2

Background

2.1 Text style transfer

[*Supervised, unsupervised, prompting, infilling (PEGF).*]

2.2 Stylometry and authorship attribution

[*to write*]

2.3 Knowledge distillation and LoRA

[*to write*]

2.4 Reinforcement learning for language models

[*RLHF/PPO, GRPO.*]

2.5 Macedonian NLP

[*CLASSLA and available resources.*]

Chapter 3

Data

3.1 The corpus

[*1,239 poems by 38 authors; 24 authors with at least 5 poems are profiled.*]

3.2 Cleaning

[*Script artifacts, poem ids.*]

3.3 Tagging and corrections

[*CLASSLA, the three normalizations, hand corrections.*]

3.4 Splits

[*987 / 126 / 126 poems, per author.*]

3.5 How little data there is

[*Poems per author; teacher examples per target.*]

Chapter 4

Stylometric foundation

4.1 Author style profiles

[*to write*]

4.2 TF-IDF palettes

[*to write*]

4.3 The semantic knowledge graph

[*to write*]

4.4 Two author classifiers

[*The reward classifier and the independent judge.*]

4.5 Validation of the pipeline

[*Tagger migration, fixed bugs, community artifact.*]

Chapter 5

The path to a teacher

5.1 Word substitution

[*to write*]

5.2 LLM infilling

[*to write*]

5.3 A scaffolded prompt for a large model

[*Which becomes the teacher.*]

Chapter 6

Method

6.1 The teacher and the synthetic data

[*to write*]

6.2 Task format

[*to write*]

6.3 Supervised fine-tuning

[*to write*]

6.4 Reward

[*Style, content, gates, weights from within-group spread.*]

6.5 GRPO

[*Merge-then-LoRA, hyperparameters, v1 and v2.*]

6.6 The second student: Gemma 4 E4B

[*to write*]

Chapter 7

Evaluation setup

7.1 Protocol

[*to write*]

7.2 Metrics

[*to write*]

7.3 The judge

[*to write*]

7.4 A decoding baseline

[*to write*]

7.5 Human evaluation

[*to write*]

Chapter 8

Results

8.1 Earlier approaches against the students

[*to write*]

8.2 Qwen3-4B: validation and test

[*to write*]

8.3 Author identification per target

[*to write*]

8.4 The decoding baseline

[*to write*]

8.5 The teacher

[*to write*]

8.6 Gemma 4 E4B against Qwen3-4B

[*to write*]

8.7 Human ratings

[*to write*]

Chapter 9

Discussion

9.1 Why the students fail

[*Data scale, style collapse, nothing for the group advantage to amplify, reward gaming.*]

9.2 The teacher's own limits

[*to write*]

9.3 Threats to validity

[*to write*]

Chapter 10

Conclusion

10.1 Future work

[*to write*]

Appendix A

Examples

A.1 One example per stage

[*to write*]

Appendix B

Prompts and hyperparameters

B.1 Teacher prompt

[*to write*]

B.2 Student prompt

[*to write*]

B.3 Training configuration

[*to write*]

Appendix C

Reproducibility

C.1 Commands

[*to write*]